Optimized Machine Learning Models and Stacking Hybrid Approach for Chronic Kidney Disease Prediction*

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Abstract. Chronic Kidney Disease (CKD) is a severe disease that requires being diagnosed at an early stage to stop potential complications. Machine learning methods show an effective solution for predicting CKD, which enables healthcare professionals to make timely decisions. However, the selection of proper models along with their performance optimization remains the key essential challenge. This study examines the effectiveness of several machine learning algorithms, such as k-nearest neighbours (KNN), decision tree classifier (DT), logistic regression, naïve bayes (NB), and gradient boosting (GB), in predicting CKD cases. In addition, a stacking hybrid model focuses on enhancing classification effectiveness as well as model robustness for preventing overfitting. Each baseline model undergoes hyperparameter tuning for maximizing accuracy through optimal setting determination. Performance evaluation of the models uses multiple measurement criteria that include accuracy, precision, recall, f1-score, Jaccard score, and AUC-ROC curve, where the optimized model and hybrid model have shown better results. These metrics provide a complete insight into how well models classify CKD and non-CKD cases correctly. The results demonstrate that each optimized model improves model performance by using hyperparameter tuning. Moreover, the stacking hybrid model (using 3 of the best optimized models) performs better than each of the optimized classifiers, showing that the potential of integrating various models can boost the ability of

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the prediction. This study highlights the optimized model and a well-tuned hybrid model can be a valuable approach for enhancing diagnostic accuracy in medical decision-making, aiding in the early identification and management of CKD.

Keywords: Chronic Kidney Disease · Machine Learning · Optimization · Hybrid Model · Healthcare.

1 Introduction

Chronic Kidney Disease (CKD) is a long-term illness in which kidney function keeps getting worse. It can progress to end-stage renal disease (ESRD) without being noticed or treated. Research from the World Health Organization (WHO) reveals that about 10% of the worldwide population struggles with CKD while millions of people stay undiagnosed because early stages show no symptoms [1]. A total of 37 million U.S. adults suffer from CKD, but only 10% receive a proper diagnosis. The main contributing factors behind CKD are diabetes and high blood pressure, which account for around 75% of all occurrences [2]. The improvement of predictive models highlights the effectiveness because early detection and intervention help minimize kidney failure risks.

In traditional methods of diagnosis, lab tests are used to measure serum creatinine levels, glomerular filtration rate (GFR), and urine protein levels [3]. However, the detection of CKD through these tests usually occurs in advanced stages where considerable kidney tissue damage has already occurred. A potential alternative is offered by machine learning algorithms that incorporate large amounts of patient data to identify patterns and estimate chronic kidney disease if it turns too severe [4, 5]. Using improved feature selection, hyperparameter tuning, and ensemble learning to make predictions more accurate and reliable will help you get even better results when you use optimized models.

This study improves CKD prediction by implementing two-step pre-processing, a hybrid stacking technique, and hyperparameter optimization procedures. Pre-processing includes filling missing values, label encoding, random sampling, variance threshold, power transformer, and the synthetic minority oversampling technique (SMOTE) technique to enhance the model performance. Grid search optimization is applied to optimize five basic machine learning models: K-Nearest Neighbours (KNN), Decision Tree (DT), Naïve Bayes (NB), Gradient Boosting (GB), and Logistic Regression (LR). The hybrid stacking model comprised the top three performers to boost prediction accuracy levels. The effectiveness of the given approach was validated by model evaluation matrices such as accuracy, precision, recall, F1-score, Jaccard score, and AUC-ROC curve. This optimized model improves the prediction of CKD with adaptability and reliability, which enables physicians to make better treatment decisions at an earlier stage.

This paper follows a structure that includes Section 2 for the literature review alongside Section 3 for the methodology description. Section 4 consists of results and analysis as well as Section 5 conclusion.

2 Literature Review

This section looks at past research on chronic kidney disease (CKD) and a number of machine learning (ML) models that can predict CKD in order to make early diagnosis and treatment better.

Ghosh et al. investigated an explainable AI-based method for early CKD forecasting using clinical data. XGBoost consistently outperformed all the ML models tested (AUC=0.9689, Accuracy=93.29%). The main added predictive attributes were creatinine, HgbA1C, and age. The aim was to improve the interpretability of the model and aid physicians in understanding the predictions made using SHAP and LIME [6]. The work of C. Chahil et al. suggested using a Kaggle-based dataset and an ensemble ML model combined with random forest and bagging to predict CKD. The model outperformed individual algorithms, having a high accuracy of 96.5%. The results showed how well ensemble learning builds CKD projections, making it useful for finding problems early on [7].

In a different study, Bilal et al. created a model for predicting CKD that uses metaheuristics to deal with issues such as computational complexity and very high dimensions. Extreme learning machines were utilized for classification, and binary gray wolf optimization was employed for feature selection. Significant preprocessing improves data quality, and assessment using different metrics demonstrates greater precision than current models. These outcomes illustrated the effectiveness of the model for improving CKD diagnosis [8]. A study by El Sherbiny et al. was developed for early CKD detection using several machine learning classifiers. A group of researchers led by El Sherbiny and others looked into a dataset of patient health using nine machine learning algorithms. They tested the dataset using four metrics: F1-score, accuracy, precision-recall, and runtime. The Adaboost model displayed the highest accuracy, 99.17%, highlighting its robustness as a viable model for predicting CKD [9].

Jongbo et al. also looked into data mining methods for predicting CKD, focusing on accuracy by picking the right features and preprocessing the data. Normalization and handling missing values were used to preprocess data. Bagging and random subspace were approached with two ensembles, and also k-nearest neighbors (KNN), naïve bayes (NB), and decision tree (DT) were applied as base classifiers. Using the UCI CKD dataset, the results were better than using ensemble methods as single classifiers, with Random Subspace on KNN giving a 100% success rate. Therefore, it was very useful for CKD diagnosis [10]. KM Almustafa et al. looked into predicting CKD using different types of classifiers, like random trees, decision tables, k-nearest neighbors, J48, and more. The outcomes showed that feature selection effectiveness worked to improve CKD diagnosis along with early detection [11].

3 Methodology

3.1 Dataset Description:

The chronic kidney disease dataset was collected from the IEEE data repository [12] for this research work. Several researchers have also used this dataset. All

data sets were gathered from a hospital in Karaikudi, India. It has 400 instances with 25 attributes and 14 nominal and 11 numeric values, as shown in Table 1, intended for classification problems. This dataset has two types of classes: CKD and non-CKD, relating to the patients without chronic kidney disease and with chronic kidney disease.

Table 1. Attribute Description Table

Attribute	**Description**	**Attribute**	**Description**
Age	Age in years	Blood Pressure	Blood pressure of the patient (mm/Hg)
Specific Gravity	Ranges of urine gravity (1.005, 1.010, 1.015, 1.020, 1.025)	Albumin	Ranges of patient albumin (0,1, 2, 3, 4, 5)
Sugar	Patient sugar range (0-5)	Red Blood Cells	RBC (Normal, Abnormal)
Pus Cell	PC (Normal, Abnormal)	Pus Cell Clumps	PCC (Present, Not present)
Bacteria	Patient bacteria (Present, Not present)	Blood Glucose Random	Patient BGR in mgs/dl
Blood Urea	Patient BU in mgs/dl	Serum Creatinine	Patient SC in mgs/dl
Sodium	Patient SOD in mEq/L	Potassium	Patient POT in mEq/L
Hemoglobin	Patient hemoglobin in gms	Packed Cell Volume	Patient pcv of % red blood cells in circulating blood
White Blood cell Count	Patient WBC in cells/cumm	Red Blood Cell Count	Patient RBC in millions/cmm
Hypertension	Patient hypertension (Yes, No)	Diabetes Mellitus	Patient diabetes mellitus (Yes, No)
Coronary Artery Disease	Patient coronary artery disease (Yes, No)	Appetite	Patient appetite (Good, Poor)
Pedal Edema	Patient edema (Yes, No)	Anemia	Patient anemia (Yes, No)
Class	Patient class (ckd, notckd)	-	-

3.2 Approach overview:

The diagram 1 presents the analysis steps for a chronic kidney disease dataset. The process begins by gathering attributes such as age, blood pressure, and sugar levels. A correlation analysis visualizes the collected data. The preprocessing phase handles three tasks: filling in missing values, encoding categorical data, and sampling. The next preprocessing steps involve a variance threshold, applying transformations alongside data balancing. The dataset consists of 80% training data and 20% testing data. Several supervised learning models, such as

k-nearest neighbours, decision trees, naïve Bayes logistic regression, and gradient boosting, are used. The hybrid model emerges from stacking different predictive models together. Optimization requires hyperparameter tuning, where the cross-validation technique ensures model reliability. Finally, performance evaluation includes accuracy but incorporates precision and recall along with the F1-score, Jaccard, and AUC-ROC curves.

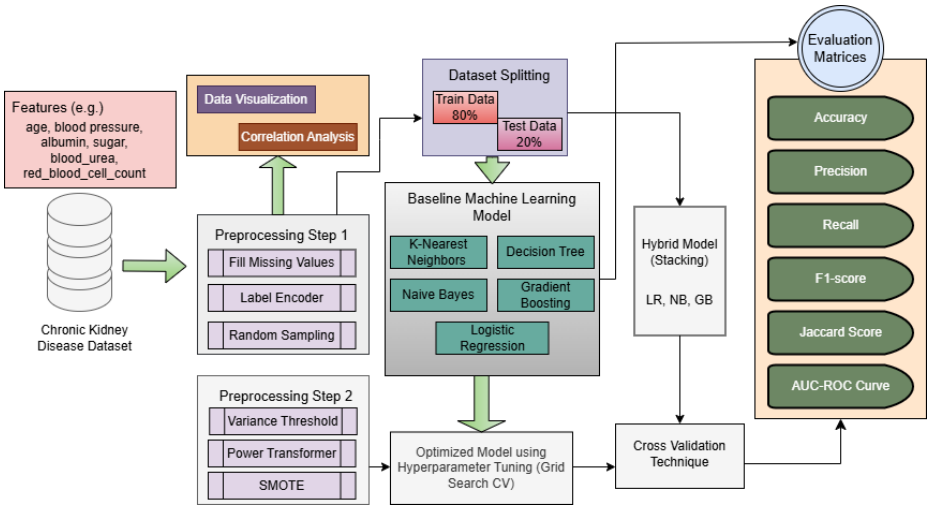

Fig. 1. Workflow of the proposed system.

3.3 Preprocessing step

The technique of data preprocessing transforms raw data into an error-free dataset. It is the fundamental phase for training every machine learning classifier algorithm. In the preprocessing phase, a systematic approach was employed to improve both the data quality and the performance of the model. The first step is handling missing values using a suitable imputation method to prevent data loss and ensure uniformity. Then, label encoding was employed for transforming categorical features into numerical values, thereby making it attainable for machine learning algorithms to deal with them effectively. A random sampling technique was used to achieve an unbiased representative dataset that minimized overfitting and improved generalization. In the second step of the pre-processing process, the optimized and hybrid models were tested with additional data modification to increase model performance. This step began with the usage of a variance threshold method for removing the low variance of features that provided less to the learning of the model. This process simplified data while removing unwanted noise from the dataset. Following that, a power

transformer technique is used for adjusting skewed data distributions to enhance model stability and convergence during training. Finally, the SMOTE provided solutions to class imbalances by creating artificial samples from minority classes, which increased model performance and generalization. This multi-step preprocessing process made it possible to make more accurate and reliable predictions for both basic and advanced models that had been trained on clean, balanced, and well-structured data.

3.4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) serves as an essential stage to understand datasets, identify patterns, and discover potential problems before starting model development. Correlation analysis and visualization methods are vital for studying data distributions and finding correlations between variables in chronic kidney disease (CKD) investigations. Correlation analysis makes it easier to determine the links between clinical indicators like creatinine levels, blood pressure, etc. The correlation matrix, or heatmap, which is shown in Figure 2, is highlighted as strongly connected with features, helping feature selection by optimizing the model and decreasing redundancy.

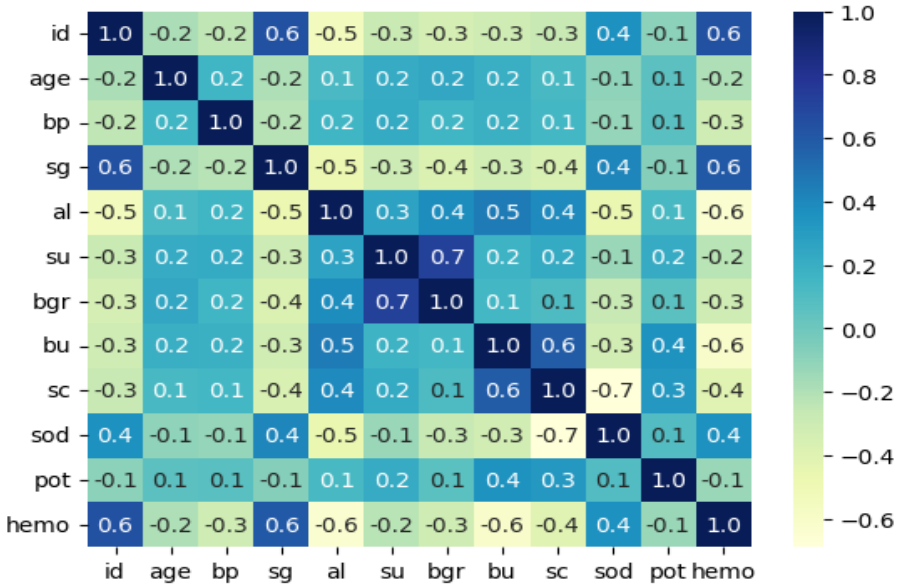

Fig. 2. Correlation matrix for numerical features on Chronic Kidney Disease (CKD) dataset.

3.5 Data Splitting

The dataset split between training and testing sets enabled proper model assessment. The training segment received 80% of the data, with testing conducted on the remaining 20%. A random state 42 was used during the split to maintain consistent and reproducible results. This approach allows models to learn diverse training samples that are evaluated on new data sets to verify reliable generalization performance.

3.6 Classification Models

A classifier model is vital for predicting disease and uses a patient's historical medical records to detect potential health issues. Initially, the classifier is employed on training data which is later tested to make target variable predictions.

Supervised Classification Models: The **K-nearest neighbours** classifier represents one of the most common data mining and machine learning applications due to its straightforward implementation and notable performance outcomes. The use of uniform k values for all test data within previous kNN approaches was demonstrated to be impractical when used in real-world scenarios [13]. A **decision tree (DT)** is a non-parametric supervised learning paradigm for classification and regression. It is widely used in machine learning applications such as medical data, pattern analysis, and so on. To generate a tree model, feature values repeatedly separate the dataset into subsets [14]. Although this algorithm is simple to interpret, it can overfit; pruning procedures like pre-pruning (max depth) and post-pruning aid in improving adaptability. **Logistic Regression** is a supervised machine learning algorithm that performs binary classification through its analysis of outcome, event, or observational probability. Logistic regression calculates the correlation between one or more independent (predictor) features and a binary dependent (outcome) [15]. A binary classifier has two types of variables, such as positive or negative. Others refer to this algorithm as logit regression, log-linear classifier, or maximum entropy classification (MaxEnt). **Gradient Boosting** is a form of ensemble supervised machine learning algorithms that leverage a final model through the combination of multiple weak learners [16]. For classification and regression tasks such as disease prediction and fraud detection, enhance the accuracy and speed of machine learning models. The probabilistic machine learning technique **Naïve Bayes** is applied for classification tasks. Based on the Bayes Theorem, it derives the assumption that attributes are conditionally independent given the class label [17]. Notwithstanding this "naïve" assumption, it is effective in a range of real-world applications such as medical diagnosis and spam detection.

Hybrid Model (LR, GB, NB): The hybrid model combines three algorithms: logistic regression (LR), gradient boosting (GB), and naïve bayes (NB). These algorithms work together to get a better classification accuracy [18]. Logistic

regression handles linear relationship analysis and remains interpretable; however, gradient boosting outperforms it by leveraging multiple iterative boosts for capturing intricate patterns. Naïve bayes implements probabilistic methods to process both categorical variables and textual data effectively. The hybrid approach eliminates overfitting and enhances predictive power by integrating these models using techniques like stacking and weighted voting. This study employs a stacking model to enhance generalization. This study employs a stacking model to enhance generalization performance. This combination is particularly beneficial in any application where a reliable and flexible classification model is required for a range of data patterns.

Algorithm 1 Hybrid Model Procedure

Input: CKD dataset
Step 1: Data pre-processing
Step 2: Split the data into training and testing sets
Step 3: Train baseline learners with optimization(Logistic Regression, Naïve Bayes, Gradient Boosting)
Step 4: Generate predictions from baseline models(Logistic prediction, Naïve prediction, Gradient prediction)
Step 5: Train meta-model using Logistic Regression
Meta-model = Train Logistic Regression(meta-features, y-valid)
Step 6: Generate final predictions
Step 7: Evaluate the model
Step 8: Predict CKD or Not CKD

3.7 Model Optimization using Hyperparameter Tuning:

Model optimization with hyperparameter tuning is vital for machine learning models to perform more effectively. Hyperparameters are configurations or settings that are set up before model training, like learning rate, tree depth, or regularization strength. Data teaches us this model parameter, which requires manual optimization for improved model performance [19]. Grid search cross-validation (gridsearchCv) is one of the most effective ways to achieve this. It extensively appears through a specified set of hyperparameters and assesses every possible combination to determine the optimum configuration. This process includes defining a range of parameters in Table 2, like max features, metric, and solver based on different models. After evaluating all possibilities, GridSearchCV returns the combination that yields the highest performance based on a selected metric like accuracy or precision. This method is advantageous because it offers an exhaustive search, provides reliable results through cross-validation, and ensures reproducibility by consistently evaluating across multiple data splits. Overall, GridSearchCV helps in finding the best hyperparameters, leading to better model generalization and accuracy.

Algorithm 2 Optimized Model for CKD Prediction

Step 1: Load CKD dataset
Step 2: Data Preprocessing (Handle missing values, Encode categorical values)
Step 3: Split data into training and testing sets
Step 4: Initialize base model
base_model = initialize_model(default_hyperparameters)
Step 5: Define hyperparameter grid
Step 6: Perform hyperparameter tuning
optimized_model = perform_hyperparameter_tuning (base_model, hyperparameter_grid, X_train, y_train, cv=5)
Step 7: Evaluate optimized model
Step 8: Predict CKD or Not CKD on unseen data

Table 2. Best Hyperparameter tuning for the optimized model.

Model	**Best Hyperparameters**
KNN	`{'algorithm': 'auto', 'leaf_size': 10, 'metric': 'manhattan', 'n_neighbors': 3, 'p': 1, 'weights': 'uniform'}`
DTC	`{'criterion': 'gini', 'max_depth': 5, 'max_features': 'log2', 'min_samples_leaf': 2, 'min_samples_split': 3, 'splitter': 'best'}`
LR	`{'C': 0.23357214690901212, 'l1_ratio': 0.0, 'max_iter': 100, 'penalty': 'l2', 'solver': 'saga'}`
GB	`{'learning_rate': 0.001, 'loss': 'exponential', 'max_depth': 5, 'max_features': 'sqrt', 'min_samples_leaf': 2, 'min_samples_split': 2, 'n_estimators': 50, 'subsample': 1.0}`
NB	`{'priors': None, 'var_smoothing': 0.1}`

3.8 Evaluation Metrics:

Measuring performance is an essential phase in developing the most accurate machine learning model. The prediction model has to be validated to ensure it corresponds to the dataset and operates effectively with undefined data. Testing model performance aims to determine accuracy levels for independent or unseen data. A variety of metrics have been determined for performance evaluation, like accuracy (Eq.1)[20], precision (Eq.2)[21], recall (Eq.3) [22], f1-score (Eq.4) [23], Jaccard score (Eq. 5) [23], and AUC-ROC curve [23].

The confusion matrix used the following metrics to evaluate classifier performance:

True Positive (TP): A true positive represents the number of correctly identified positive class instances.

True Negative (TN): A true negative shows the correct classification rate of negative observations.

False Positive (FP): The number of incorrect positive class identifications.

False Negative (FN): The count of incorrectly identified negative samples.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

$$\text{F1-score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} \tag{4}$$

$$J(y, Y) = \frac{|y \cap Y|}{|y \cup Y|} \tag{5}$$

4 Result and Analyssis

This section presents the performance metrics of each classifier, including the confusion matrix, accuracy, precision, recall, f1-score, Jaccard score, and roc-curve. The Jupyter Notebook was applied to carry out the tests using the Python 3.14 programming language on an Intel Core i7 13th generation processor and 16 GB RAM. Furthermore, this study implemented an optimized model for the measurement of CKD by tuning hyperparameters in each classifier's algorithm.

4.1 Confusion Matrix

Table highlights the values of the confusion matrix for the applicable classifiers, as stated earlier, as well as the case study of logistic regression (LR). Table 3 shows that the TP of the 48 cases that were categorized as positive (60%) was 4, and the FP was 4 for positive cases that were labeled as negative. This was thought to be the 5% error in the categorization of positive cases. For the number of negative cases that were classified as negative cases (31.25%), the TN is 25 and the FN is 3 (3.75% error), which classifies as positive cases for negative cases. On the other hand, Table 4 shows the values of the confusion matrix for the optimized and hybrid models mentioned previously, taking the hybrid model scenario into account. Table 4 displays 49 instances of true positives (TP), labeled as positive cases (49%). The false positive case was labeled as an FP, which meant that there was only a 1% error in the classification. The true positive (TN) category includes 50 cases where the false negative (FN) indicates no error in the classification of negative cases.

Table 3. Confusion Matrix for the baseline Models

Model	TP	FP	FN	TN
KNN	38	14	11	17
DTC	52	0	4	24
LR	48	4	3	25
GB	52	0	2	26
NB	51	1	2	26

Table 4. Confusion Matrix for the optimized Models

Model	TP	FP	FN	TN
KNN	46	4	0	50
DTC	48	2	0	50
LR	49	1	0	50
GB	48	2	0	50
NB	47	3	0	50
Hybrid Model	49	1	0	50

4.2 Classification models on training and testing performance (Baseline)

The table 5 shows the performance metrics of different machine learning models that operate on training data. The study looks at five performance metrics: accuracy, precision, recall, f1-score, and Jaccard score. These are kNN, DTC, LR, GB, and NB. On the training set, both DTC and GB models were 100% accurate across all metrics. This means they were perfect at classifying the data. NB exhibited good performance, with high precision, recall, and an accuracy of 96.87%. LR model delivered strong outcomes with 91.56% accuracy and 86.25% precision in its yield. The KNN model had the lowest accuracy (82.19%), with a relatively lower precision (71.81%) and Jaccard score (65.24%), though it maintained a good recall value (87.70%). The tree-based models demonstrated superior performance overall compared to KNN, which had minimal effective classification results. In addition, Figure 3 shows the classification performance as a bar chart that compares the training models that were looked at.

Table 5. Training Data Performance Metrics of Baseline Machine Learning Algorithms

ML models	Accuracy	Precision	Recall	F1-score	Jaccard score
K-nearest neighbors	82.19%	71.81%	87.70%	78.97%	65.24%
Decision tree	100%	100%	100%	100%	100%
Logistic regression	91.56%	86.25%	92.62%	89.32%	80.71%
Gradient boosting	100%	100%	100%	100%	100%
Naive Bayes	96.87%	92.42%	100%	96.06%	92.42%

Table 6 displays the results of testing performance metrics from various machine learning models, including accuracy, precision, recall, f1-score, and jaccard score. The naïve Bayes and gradient boosting methods got the best results, with an accuracy rate of 97.50%. They also got the best scores for precision, recall, and an F1 score. The Decision Tree model also showed strong performance, with 95.00% accuracy and perfect precision (100%), but a slightly lower recall (85.71%). Logistic regression performed well, maintaining 91.25% accuracy, while K-Nearest Neighbors (KNN) had the lowest performance, with 68.75% accuracy

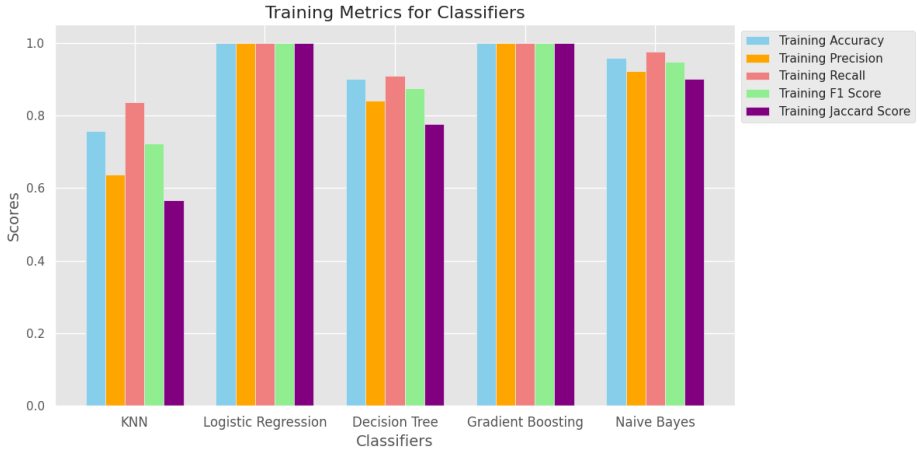

Fig. 3. Evaluation metrics for training data on baseline models

and the lowest Jaccard score (40.48%), indicating weaker classification ability. Figure 4 graphical comparison shows that tree-based models outperformed KNN in every evaluation metric.

Table 6. Testing Data Performance Metrics of Baseline Machine Learning Algorithms

ML models	Accuracy	Precision	Recall	F1-score	Jaccard score
K-nearest neighbors	68.75%	54.84%	60.71%	57.63%	40.48%
Decision tree	95.00%	100%	85.71%	92.30%	85.71%
Logistic regression	91.25%	86.20%	89.28%	87.71%	78.12%
Gradient boosting	97.50%	100%	92.85%	96.29%	92.85%
Naive Bayes	97.50%	96.29%	92.85%	94.54%	89.65%

4.3 Optimized Classifier Models performance

The performance analysis of optimized classifier algorithms appears in Table 7, with kNN, DTC, LR, GB, NB, and a hybrid model (LR, NB, GB) all included. The hybrid model reached the highest performance among all models, with 99% accuracy, precision, F1-score, and Jaccard score. This exhibits its ability to classify effectively. LR model also performed exceptionally well, matching the hybrid model in all metrics except accuracy (99.00%). DTC and GB both reached 98.00% accuracy, demonstrating strong classification ability. NB showed slightly lower accuracy (97.00%) but maintained a perfect recall of 100%, making it useful for minimizing false negatives. The accuracy of kNN reached 96.00% and proved better than previous results despite being the least accurate method among all implementations. Figure 5 illustrates the visual relationship, which

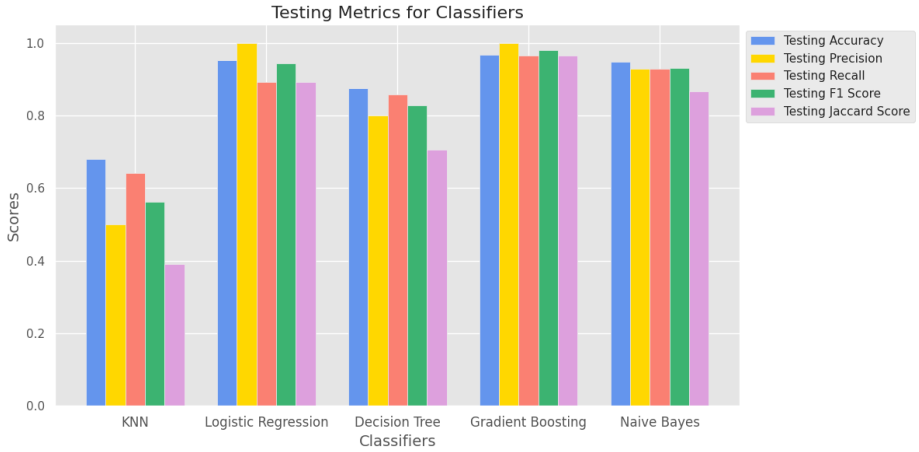

Fig. 4. Evaluation metrics for testing data on baseline models

proves the hybrid model outperforms other models in optimizing their classification results.

Table 7. Testing Data Performance Metrics of Optimized Machine Learning Algorithms and Hybrid Model

ML models	Accuracy	Precision	Recall	F1-score	Jaccard score
K-nearest neighbors	96.00%	92.59%	100%	96.15%	92.59%
Decision tree	98.00%	96.15%	100%	98.03%	96.15%
Logistic regression	99.00%	98.03%	100%	99.00%	98.03%
Gradient boosting	98.00 %	96.15%	100%	98.04%	96.15%
Naive Bayes	97.00%	94.33%	100%	97.08%	94.33%
Hybrid Model	99.00%	98.03%	100%	99.00%	98.03%

4.4 Roc-curve:

Figure 6 shows the Receiver Operating Characteristics (ROC) curve for various machine learning models that are used to guess kidney disease. To see how well the models can classify things, the graph shows the relationship between sensitivity (the rate of true positives) and specificity (the rate of false positives). The DTC and GB models achieved impressive ROC scores of 0.95, which demonstrates excellent classification abilities. The ROC value of NB reached 0.92, while the LR model showed a moderate ROC score of 0.89. However, kNN achieved the worst results with a ROC score of 0.67, indicating its poor discriminatory capacity. Figure 7 displays another ROC curve, which incorporates both optimized models and a hybrid model for the prediction of kidney disease. The graph also compares various classifiers by sensitivity versus 1 specificity. The hybrid model achieved the best performance in the ROC score (0.98), which was also equal to

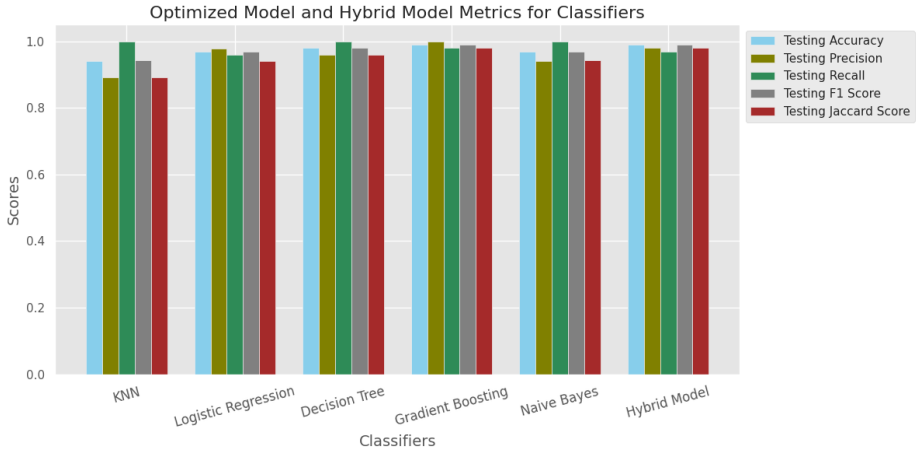

Fig. 5. Evaluation metrics for testing data on optimized models and hybrid model.

LR and GB. The performance of NB and DTC matched each other at 0.97 ROC, but k-NN demonstrated lower performance with 0.94 ROC. The hybrid model outperformed all other optimized models and proved most useful for detecting kidney disease based on its excellent performance measurements.

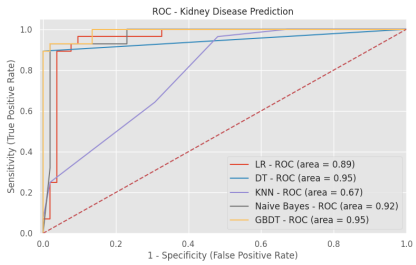

Fig. 6. AUC-ROC curve for baseline models.

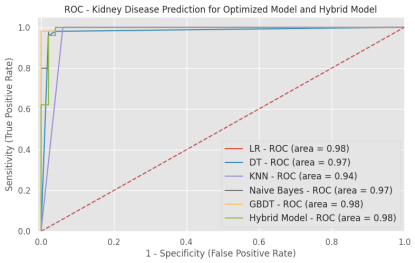

Fig. 7. AUC-ROC curve for optimized models and hybrid model.

4.5 Discussion

Table 8 provides a comparison of several studies on CKD prediction alongside our proposed method using machine learning. N. A. Almansour et al. achieved the highest accuracy of 98.75% using SVM and ANN on the UCI CKD-1 dataset with K-fold cross-validation. O.A. Jonbo et al. reported 95% accuracy using Naive Bayes (NB) and Decision Trees (DT), with 70% of the data for training and 10% for testing. Ghosh et al. used multiple algorithms (LR, RF, NB, XGBoost)

on hospital data from Al-Ain City, UAE, and achieved an accuracy of 93.29%. C. Chahil et al. reached 96.5% accuracy with a combination of LR, RF, and voting models on the Kaggle dataset. Bilal et al. achieved 96.98% accuracy using AdaBoost, KNN, and SVM on the UCI dataset with an 80/20 training-test split. Our proposed approach achieved the highest accuracy of 99% by combining KNN, DT, GB, NB, and LR using K-fold cross-validation. These results indicate that while individual models perform well, hybrid approaches tend to deliver the best accuracy. Future research should explore ensemble methods, more diverse datasets, and stronger validation techniques to improve model generalization and performance. Additionally, comparing different validation methods like K-fold cross-validation and simple train-test splits highlights the importance of using robust validation strategies to reduce overfitting and increase prediction reliability.

Table 8. Comparative Analysis with Recent Studies for CKD Prediction

Author	**Data Source**	**Algorithm**	**Validation**	**Experimental Environment**	**Accuracy (%)**
N. A. Almansour et al.	UCI ML Repository dataset CKD-1	SVM, ANN	K-fold Cross Validation (k=10), Training set: 90% Test set: 10%	WEKA	98.75
O.A. Jonbo et al.	UCI ML Repository	NB, DT	Training set: 70%, Test set: 10%	Python 3.3	95.00
Ghosh et al.	Tawan Hospital in Al-Ain City, Abudhabi, UAE	LR, RF, DT, NB, XGBoost	K-fold Cross Validation (K=10)	Python 3.3	93.29
C. Chahil et al.	Kaggle	LR, GP, Bagging, RF, Vote	-	Python	96.5
Bilal et al.	UCI Dataset	AdaBoost, KNN, NB, SVM, ELM	Training set: 80%, Test set: 20%	Matlab	96.98
Proposed Approach	IEEE CKD Dataset	KNN, DT, GB, NB, LR, Stacking Hybrid	K-fold Cross Validation (k=5), Training set: 80%, Test set: 20%	Python 3.14	99.00

5 Conclusion

The extremely high elevation of chronic kidney disease in today's world is observed as an imminent threat to human beings. This paper highlights five basic machine learning models, an optimization model on base learners, and a stacking hybrid model along with three best-optimized models for enhancing their precision of classifying for effective chronic kidney disease diagnosis. The accuracy of a model is highly significant because it is directly involved with human lives. The use of multiple hyperparameter tuning for optimization models as a method to produce results from different machine learning models, including K-NN, DTC, GB, LR, and NB, displayed their effectiveness on the chronic kidney disease dataset through the method of improved computer-aided diagnostic model classification accuracy. The hybrid stacking model is also used by integrating three best-optimized models for model robustness and reliability and could

get better results in classifying chronic kidney disease. The provided model's performance was determined, and comparisons between the baseline model and optimized model as well as the stacking hybrid model were carried out. According to the findings from experimental results, the developed optimized model's approach to detecting chronic kidney disease performed more effectively than baseline classification models. The hybrid model also showed a better result with trustworthiness, robustness, and adaptability. The analysis's evident outcomes demonstrated that the optimized method and hybrid model are more precise than baseline model classifiers, therefore making them suitable for predicting chronic kidney disease.

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